SMART INDIA HACKATHON 2025

TEAM ECOX

Problem Statement ID – SIH25122

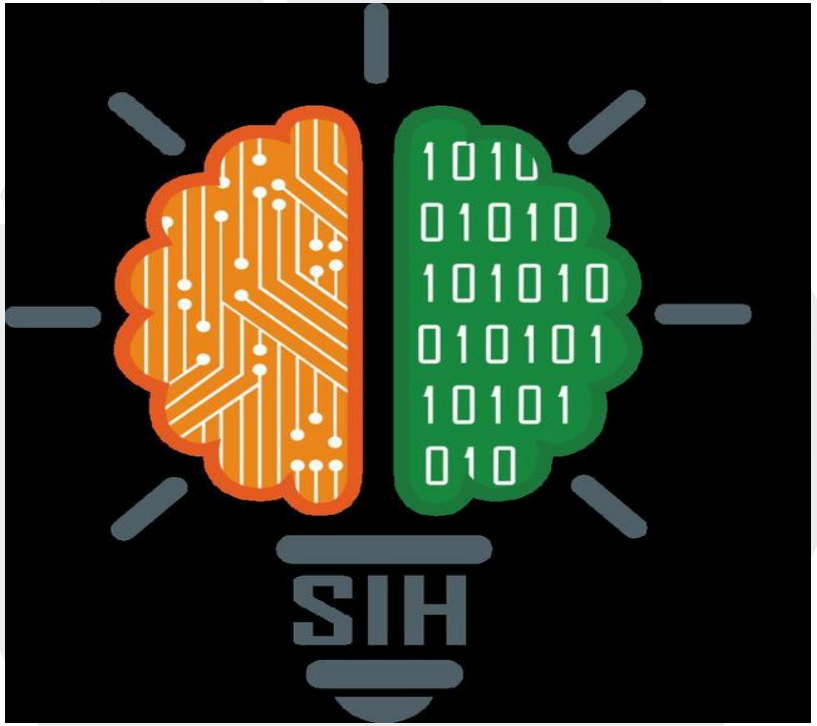
Problem Statement Title – Student Innovation

Theme – Disaster Management

PS Category – Hardware

Team ID – 80643

Team Name – Team EcoX



Team EcoX

IDEA TITLE

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PROBLEM STATEMENT:

Lack of affordable, portable, and real-time solutions for monitoring harmful emissions and ensuring protection from toxic gases remains a major challenge. Uncontrolled **emissions from industries, mines, and urban areas** expose workers to toxic gases compromising their safety as well as pollute the environment worsening inhabitants health. Conventional monitoring systems are **costly and complex,** leaving two critical gaps: **Worker Safety and Pollution Control**.

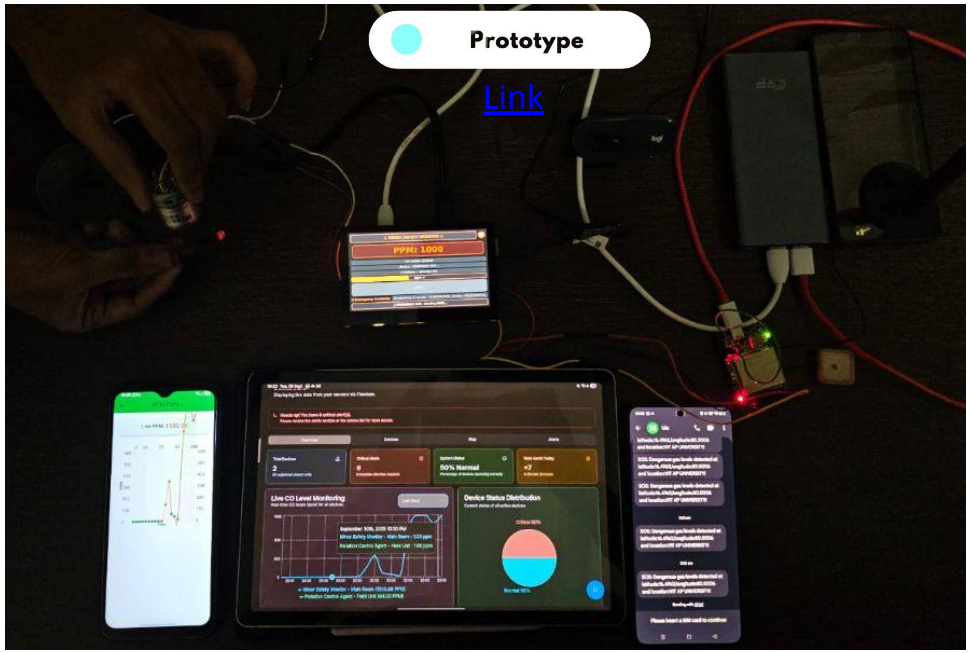
UVP:

A portable and multi-purpose IoT-based emission monitoring device that combines **real-time tracking, geo-tagged data, predictive analytics, and instant safety alerts.** Our solution combines **worker protection, and public health impact in one scalable and low-cost system,** making **worker safety and pollution control accessible to all mines, plants, and industries,** especially small and medium enterprises, unlike traditional costly and complex CEMS solutions.

SOLUTION:

Our solution is a **multi-purpose, affordable emission monitoring system** designed to improve **worker safety** and **pollution control**:

- **IoT Device:** Portable, real-time emission monitoring for toxic gases in industries, mines, and factories.
- **Dashboard:** Centralized platform with analytics, insights, and predictive alerts for industries and government agencies.
- **Camera Monitoring:** Visual incident documentation and remote monitoring for comprehensive safety oversight.
- **Safety Alerts:** A speaker is integrated to provide loud sound alerts, while multi-channel notifications via the hardware (using LTE/VoLTE, Wi-Fi, and Bluetooth) ensure immediate response to leaks or emission violations.

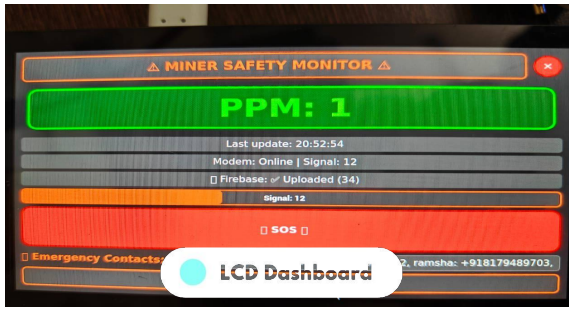


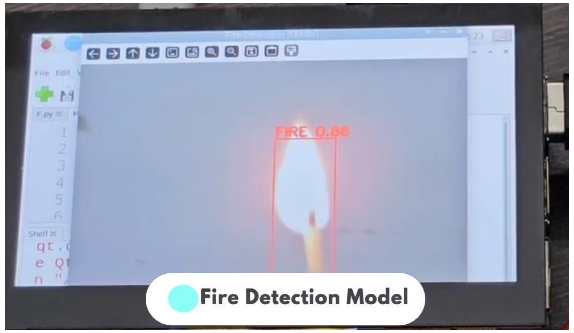
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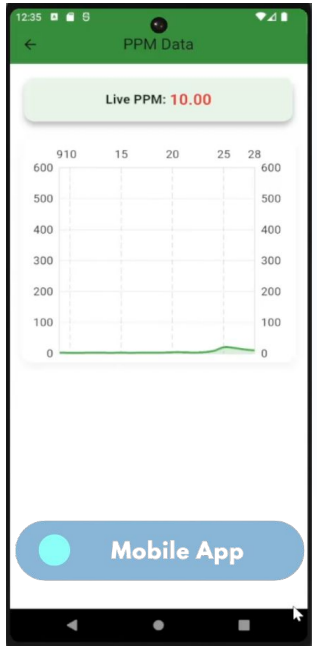
TECHNICAL APPROACH

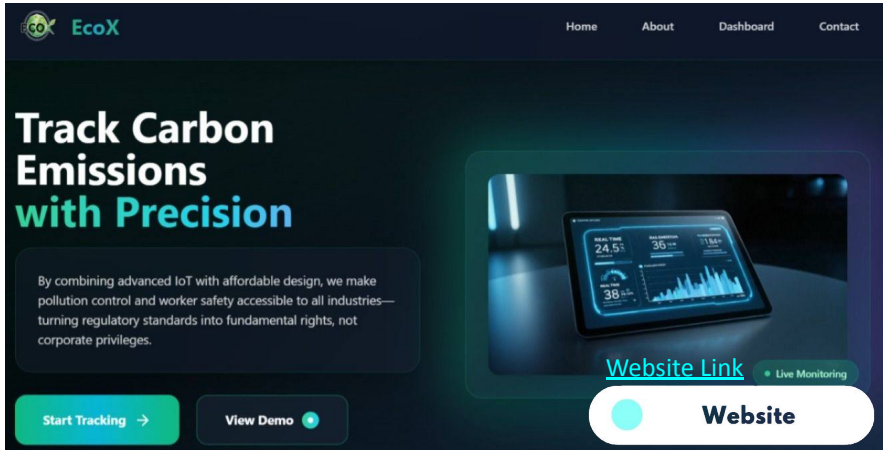
Emission Monitoring & Alert System

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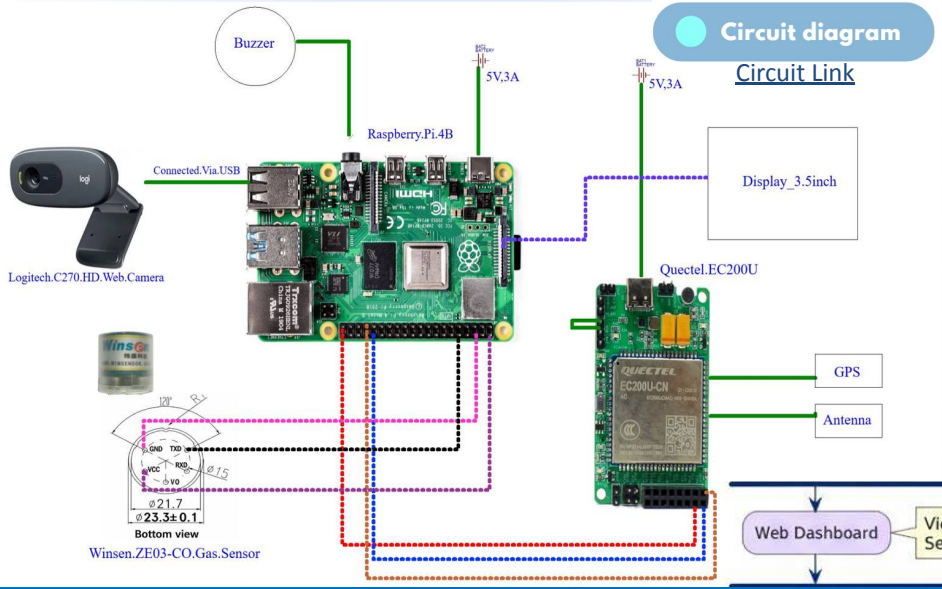
Hardware & Software Layer

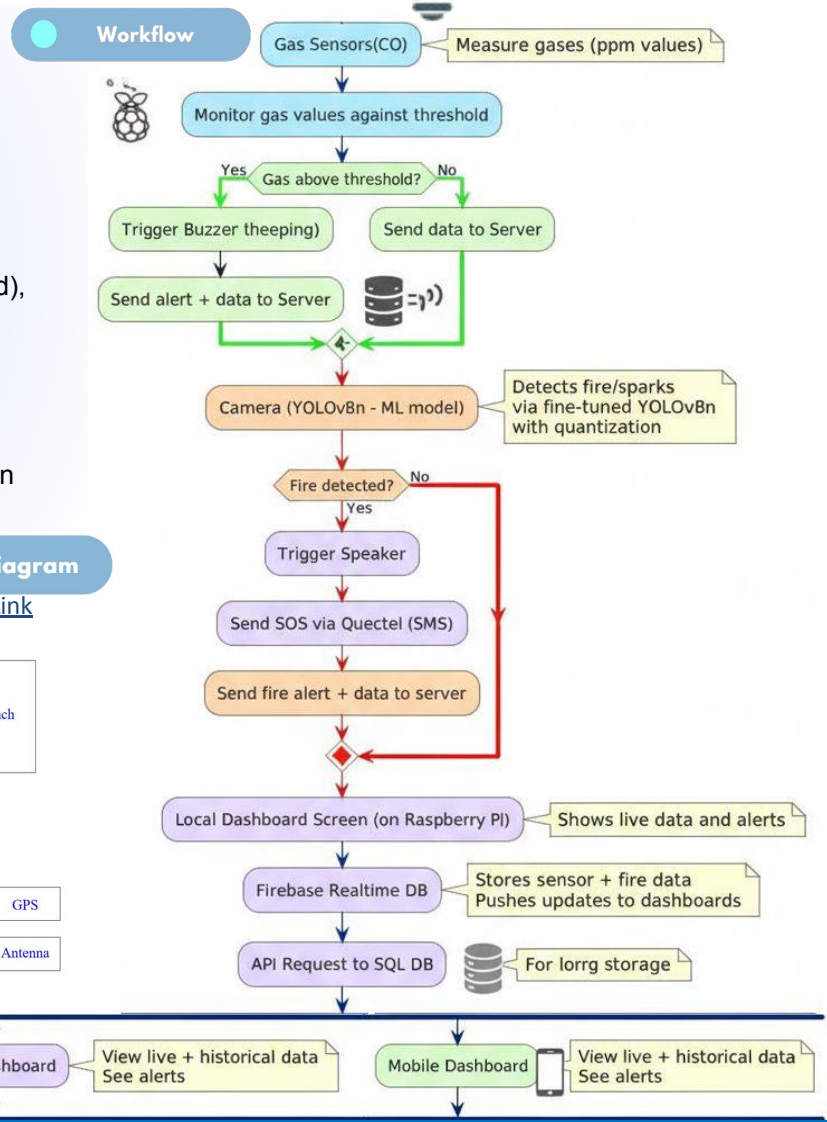
Hardware

Raspberry Pi 4B,
Winsen ZEO3-CO Gas Sensor,
Quectel EC200UCNAA 4G+GPS,
WaveShare 4.3" Touch Display,
Logitech Camera,
PAM 8403 Amplifier,
8 ohm Speaker
Battery/Powerbank.

Software

Server:
Firebase API + SQL
Dashboards:
Local (device),
Website (React.js/Next.js/Tailwind),
Mobile (Flutter) for
Backend:
Node.js + TypeScript/JavaScript
AI/ML:
YOLOv8n on Pi for Fire Detection
(Float16 for quantization)





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FEASIBILITY AND VIABILITY

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Feasibility

- **Low-Cost Hardware:** Uses commercially available, reliable components.
- **Portable & Modular:** Lightweight design with detachable sensors.
- **Scalable:** Real-time monitoring via a robust software stack.
- **Battery-Powered:** Ensures flexible deployment.

Scalability

- **Affordable Monitoring:** Enterprise-grade capability at just ₹12,000–₹15,000.
- **Massive Savings:** CEMS cost ₹1.2–₹13 lakh, making our solution up to 90% cheaper.
- **SME-Friendly:** Accessible to Small and Upcoming Enterprises who cannot afford CEMS.
- **Scalable Deployment:** Low per-unit cost enables wide rollouts across factories or sites.

Reliability

- **Proven Hardware:** Uses reliable, industry-tested components.(Winsen ZE03 CO sensor certified for industrial use,Quectal EC200U-CNAA, which has licenses in multiple region, used in industries)
- **Fail-Safe:** Stores data on a Class 10 SD card and auto-syncs when connectivity returns.
- **Consistent:** Easy calibration and replaceable sensors.
- **Scalable:** Portable for diverse industries.

No Fault Tolerance

- **Reliable Data Capture:** Continues recording even if the cloud connection is lost.
- **Local Storage:** Caches data securely on SD card.
- **Automatic Sync:** Data uploads automatically once connectivity is restored.
- **Uninterrupted Monitoring:** Ensures real-time and historical records are never lost.

Challenges and Risks

- **Sensor calibration** for accuracy in different industrial environments.
- **Battery life** and power optimization for continuous monitoring.
- **Data reliability & latency** in remote areas with poor connectivity.
- Integration of **camera monitoring & ML models** is still in development.

Strategies to overcome

- **Sensor Calibration:** Regular protocols, reference sensors, AI drift correction.
- **Power Management:** Low-power hardware, sensor sleep cycles, solar/long-life batteries.
- **Connectivity:** Multi-network fallback, local buffering, delayed sync.
- **Environmental Robustness:** Rugged IP-rated enclosures, heat/dust protection, field-tested.

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IMPACT AND BENEFITS

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Key Features

GPS & Tracking: Precise geolocation for emission source mapping.

Modular Design: Lightweight and portable for easy deployment in diverse environments.

Detachable Sensors: Flexible sensor configuration for customization to different gas types.

Interactive Dashboard: Intuitive touch screen for real-time data visualization and control.

Connectivity (LTE / VoLTE / Bluetooth / WiFi): Ensures continuous monitoring and alerts for critical safety information.

Low-Cost & Scalable: Affordable manufacturing for accessibility by small and medium enterprises.

Potential Impact

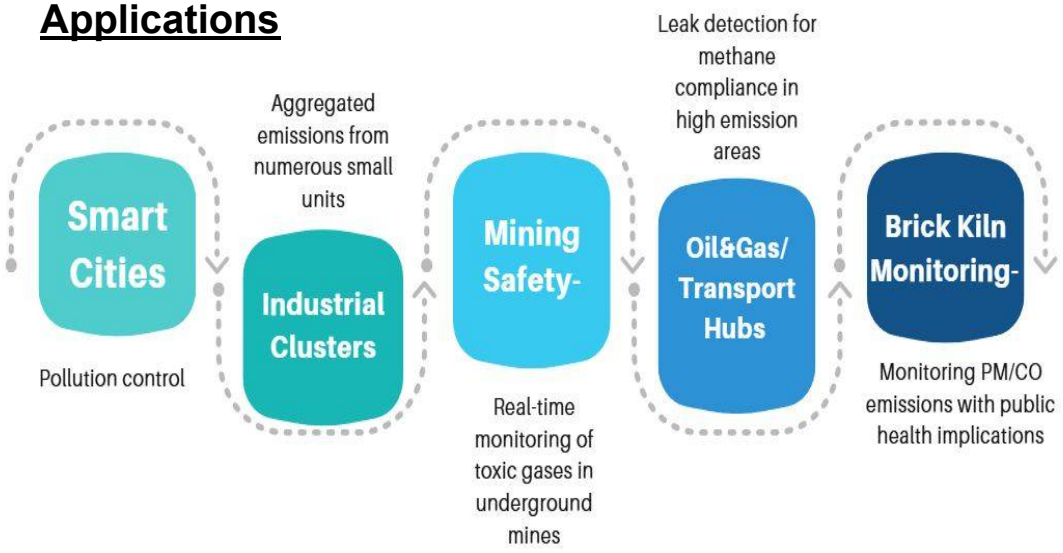
Industries & Factories: Affordable monitoring, compliance made easy, safer workplaces.

Workers: Real-time alerts, reduced health risks, trust gain.

Government & Regulators: Access to real-time geo-tagged data, simplified enforcement.

Communities/Public: Cleaner air, fewer health issues, improved quality of life.

Applications



Benefits of the Solution

Social:
Real-time alerts keep workers safe. Protects nearby communities from harmful emissions.

Economic:
Cost-effective alternative to CEMS. Avoids fines & compensation costs. Enables predictive maintenance.

Environmental:
Cleaner air & reduced emissions. Supports sustainable practices.

Technological:
IoT + AI integration for real-world impact. Scalable for smart city and urban air monitoring.

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RESEARCH AND REFERENCES

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IoT & Pollution Monitoring

Praveenchandar, J., Vetrithangam, D., Kaliappan, S., Karthick, M., Pegada, N. K., Patil, P. P., Rao, S. G., & Umar, S. (2022). IoT-Based Harmful Toxic Gases Monitoring and Fault Detection on the Sensor Dataset Using Deep Learning Techniques. Scientific Programming, 2022. https://doi.org/10.1155/2022/7516328

Worker Safety & Industrial Monitoring

Raj, M. P., & Sairam, R. (2019). *IoT Based Smart Helmet for Hazardous Environment Worker Safety*. International Journal of Recent Technology and Engineering, 8(2S11), 120–124.

IoT-Based Harmful Toxic Gases Monitoring and Fault Detection on the Sensor Dataset Using Deep Learning Techniques. Scientific Programming, 2022. https://onlinelibrary.wiley.com/doi/abs/10.1155/2022/7516328

IoT in Environmental Compliance

World Health Organization (WHO). (2018). *Ambient (Outdoor) Air Quality and Health*. Retrieved from: https://www.who.int

How India Can Adopt Innovative Tech To Cut Industrial GHG Emissions. Retrieved from: https://www.forbes.com/sites/ankitmishra/2024/09/08/how-india-can-adopt-innovative-tech-to-cut-industrial-ghg-emissions/

Sensor & Hardware Documentation

Winsen Electronics Technology Co., Ltd. (2023). *MQ Series Gas Sensors Datasheet*. Available at: http://www.winsen-sensor.com

Raspberry Pi Foundation. (2023). *Raspberry Pi 4 Model B Documentation*. Available at: https://www.raspberrypi.com/documentation

Related Standards

Central Pollution Control Board (CPCB), Government of India. (2020). *Continuous Emission Monitoring Systems (CEMS) Guidelines*. Retrieved from: https://cpcb.nic.in

International Labour Organization (ILO). *Workplace Safety and Health in Hazardous Industries*. Available at: https://www.ilo.org

Please Refer to this link for extended references and details about project Documents